Problem 15.25

To accumulate 8000 at the end of $3n$ years, deposits of 98 are made at the end of each of the first n years and 196 at the end of each of the next $2n$ years. The annual effective rate of interest is i. You are given $(1+i)^n = 2.0$. Determine i.

Solution

Let

$$(1+i)^n = 2.$$

The deposits of 98 during the first n years accumulate to time n as

$$98s_{\overline{n}|},$$

and then accumulate for another $2n$ years to time $3n$. Since

$$(1+i)^{2n} = ((1+i)^n)^2 = 2^2 = 4,$$

the accumulated value at time $3n$ of the first block of deposits is

$$98s_{\overline{n}|}(1+i)^{2n} = 98s_{\overline{n}|}(4).$$

The deposits of 196 during the next $2n$ years form an annuity-immediate of length $2n$, so their accumulated value at time $3n$ is

$$196s_{\overline{2n}|}.$$

Thus,

$$98s_{\overline{n}|}(4) + 196s_{\overline{2n}|} = 8000.$$

Using

$$s_{\overline{m}|} = \frac{(1+i)^m - 1}{i},$$

we get

$$98\left(\frac{(1+i)^n - 1}{i}\right)(4) + 196\left(\frac{(1+i)^{2n} - 1}{i}\right) = 8000.$$

Substitute $(1+i)^n = 2$ and $(1+i)^{2n} = 4$:

$$98\left(\frac{2-1}{i}\right)(4) + 196\left(\frac{4-1}{i}\right) = 8000.$$

Therefore,

$$\frac{392}{i} + \frac{588}{i} = 8000.$$

So

$$\frac{980}{i} = 8000.$$

Solving for i,

$$i = \frac{980}{8000} = 0.1225.$$

Hence,

$$\boxed{i = 12.25\%}.$$